LESSON 3.7

Ratio Reasoning: Convert Measurements Between Systems

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can use ratio reasoning to convert measurements between the customary and metric systems.

WHY THIS MATTERS The academy's kitchen orders from suppliers in two different countries. One ships in pounds and quarts, the other in kilograms and liters. Compare measurements by converting with the appropriate factor. Some factors are exact; rounded factors give approximate results. One supplier ships 10 pounds of rice and the other ships 5 kilograms. Use 1 pound ≈ 0.45 kilogram to find which shipment is more.

REMEMBER

Ratio Reasoning: Convert Measurements Between Systems

Converting Measurements Between Systems Kilometers and miles both measure distance, but they come from different measurement systems, so the numbers cannot be compared directly. A conversion factor links the two systems, and a ratio table scales it up to the amount you need.

1. Identify the conversion ratio connecting systems (e.g. 1 in = 2.54 cm, 1 kg ≈ 2.2 lbs).
2. Set up a ratio table or proportion aligning customary and metric units.
3. Multiply or divide by the conversion factor to compute the equivalent measurement

Words: **Conversion factor *(Factor de conversión)*** **Customary system *(Sistema usual)*** **Metric system *(Sistema métrico)*** **Approximately *(Aproximadamente)*** **Ratio table *(Tabla de razones)***

1 **Circle the letter of the best answer. Show how you know.**

The other supplier sells flour by the kilogram. The academy kitchen needs 22 pounds of flour. Using 1 kg ≈ 2.2 lb, about how many kilograms should it order?

(A) About 48.4 kg
(B) About 19.8 kg
(C) About 100 kg
(D) About 10 kg

HINT Write the ratio 1 kg : 2.2 lb.

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

A recipe from the U.S. supplier uses 3 cups of milk per batch. The academy buys milk in liters, and 1 cup ≈ 0.24 liter. About how many liters of milk do 5 batches need?

(A) About 3.6 liters
(B) About 0.72 liter
(C) About 62.5 liters
(D) About 36 liters

HINT How many cups do 5 batches need?

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

How many cups are in 3 quarts? (1 quart = 4 cups)

(A) 3/4 cup
(B) 7 cups
(C) 12 cups
(D) 16 cups

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A rope is 240 centimetres long. How many metres is that? (1 m = 100 cm)

(A) 0.24 m
(B) 2.4 m
(C) 24 m
(D) 240 m

My work — one step per line	What I did to both sides

1. **Look** *What is done to the variable?*
2. **Undo** *Do the opposite — both sides.*
3. **Simplify** *One side at a time.*
4. **Check** *Put it back in.*

5 **Circle the letter of the best answer. Show how you know.**

Which conversion factor changes feet into inches?

(A) 1 ft / 12 in
(B) 1 in / 12 ft
(C) 3 ft / 1 yd
(D) 12 in / 1 ft

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

A jug holds 5 litres. How many millilitres is that? (1 L = 1,000 mL)

(A) 0.005 mL
(B) 50 mL
(C) 500 mL
(D) 5,000 mL

My work — one step per line	What I did to both sides

1. **Look** *What is done to the variable?*
2. **Undo** *Do the opposite — both sides.*
3. **Simplify** *One side at a time.*
4. **Check** *Put it back in.*

EXPLAIN YOUR THINKING

Why does 1 inch = 2.54 cm use =, while 1 mile ≈ 1.609 km uses ≈? What does the rounding change?

SENTENCE STARTER The inch factor is ___ by definition. The mile factor shown is ___, so I write ___ for its result.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each customary unit with its metric equivalent. The inch conversion is exact; the other factors shown are rounded.

_____ 1 inch	A 2.54 centimeters
_____ 1 mile	B 1.609 kilometers
_____ 1 pound	C 0.454 kilogram
_____ 1 United States liquid quart	D 0.946 liter

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

Evelyn is driving in Canada. The highway sign reads 100 kilometers per hour. She is used to speed limits in miles per hour, and 1 kilometer is about 0.6 mile. Using 1 km ≈ 0.6 mi, about how many miles is 10 kilometers?

A 0.06 mile
B 6 miles
C 16 miles
D 60 miles

PER 1

3 **Circle the letter of the best answer. Show how you know.**

Evelyn is driving in Canada. The highway sign reads 100 kilometers per hour. She is used to speed limits in miles per hour, and 1 kilometer is about 0.6 mile. Using 1 lb ≈ 0.45 kg, about how many kilograms is a 30-pound dog?

(A) 4.5 kg
(B) 13.5 kg
(C) 30.45 kg
(D) 66 kg

PER 1	

4 **Circle the letter of the best answer. Show how you know.**

Evelyn is driving in Canada. The highway sign reads 100 kilometers per hour. She is used to speed limits in miles per hour, and 1 kilometer is about 0.6 mile. Why does a conversion using the rounded factor 1 mile ≈ 1.609 km use ≈?

(A) Because metric units are always bigger
(B) Because the amounts are actually different
(C) Because ratios cannot be exact
(D) The conversion factors are rounded approximations

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

One inch equals exactly 2.54 cm. How many centimeters are in 4 inches?

(A) 1.57 cm
(B) 6.54 cm
(C) 10.16 cm
(D) 254 cm

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

Which unit is metric?

(A) Quart
(B) Liter
(C) Gallon
(D) Pint

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Why does 1 inch = 2.54 cm use =, while 1 mile ≈ 1.609 km uses ≈? What does the rounding change?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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